

Homework — 2.1.4 Thinking Logically

Name: _____ Date: _ **Mark:** ____ / 20

OCR H446 — set after the 2.1.4 lesson.

Part A — This week's topic (~14 marks)

1. A theme-park ride allows a rider on only if they are **at least 1.4 m tall**.

(a) Write the exact Boolean **condition** tested at this decision point. **[1]** (AO2)

(b) State the outcome of the TRUE branch and the outcome of the FALSE branch. **[2]** (AO2)

2. A bank allows a cash-machine withdrawal only if the amount requested is **no more than** the account balance **and** no more than the daily limit.

Write a suitable Boolean condition using a logical operator, and state what happens when the condition is **false**. **[2]** (AO2)

3. A water-tank controller turns a pump **ON** when the level falls below 20%, and **OFF** when it rises above 80%; otherwise it leaves the pump unchanged.

(a) Write the two Boolean conditions used at these decision points. **[2]** (AO2)

(b) State what the controller does when the level is exactly 50%. **[1]** (AO2)

4. A shop gives a loyalty discount when a customer has spent **£100 or more**. A student writes the condition `spent > 100`.

State whether a customer who has spent exactly £100 receives the discount under this condition, and rewrite the condition so that exactly £100 **does** qualify. **[2]** (AO2)

5. An online exam system decides a candidate's grade band. If the score is **70 or above** it awards a Distinction; otherwise, if the score is **40 or above** it awards a Pass; otherwise it awards a Fail.

Identify the decision points, write the exact Boolean conditions in the correct order, and state the outcome of each branch. **[4]** (AO2)

Part B — Retrieval: keep it fresh (~6 marks)

6. (2.1.3) In a payroll program the order of steps is: read hours → calculate gross pay → deduct tax → output payslip. Explain why "deduct tax" must come **after** "calculate gross pay". **[2]** (AO2)

7. (1.3.2) State what is meant by a **primary key** and a **foreign key** in a relational database. [2] (AO1)

8. (1.4.1) Convert the denary number **45** into **8-bit binary**. [2] (AO2)

